



# Arithmetic Operators Basics

## Brief Overview

This note covers Arithmetic Operators and was created from a [4-page PDF](#). It explains how to use the four basic arithmetic signs, the % modulo operator, operator precedence rules, best practices for parentheses, and a fun dog-to-human age conversion example.

## Key Points

- **Basic arithmetic operations**: addition, subtraction, multiplication, division
  - **Modulo operator**: returns a remainder, useful for even/odd checks
  - Operator precedence: \* / % before + - unless parentheses override it
  - Practical example: converting a dog's age to human years with a clear formula
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## Arithmetic Operators +

### Basic Operators

- **Addition** – use the plus sign +

$a + b$  adds  $a$  and  $b$ .

- **Subtraction** – use the minus sign -

$a - b$  subtracts  $b$  from  $a$ .

- **Multiplication** – use the asterisk \*

$a * b$  multiplies  $a$  by  $b$ .

- **Division** – use the forward slash /

$a / b$  divides  $a$  by  $b$ .

### Modulo Operator %

**Definition** – The **modulo** operator % returns the *remainder* after dividing one integer by another.

| Expression   | Calculation               | Remainder (Result) |
|--------------|---------------------------|--------------------|
| $9 \bmod 6$  | $9 \div 6 = 1$ remainder  | 3                  |
| $6 \bmod 4$  | $6 \div 4 = 1$ remainder  | 2                  |
| $12 \bmod 8$ | $12 \div 8 = 1$ remainder | 4                  |

## Practical Uses of Modulo

- **Even/Odd Test** – A number  $n$  is **even** if  $n \bmod 2 = 0$ ; otherwise it is **odd**.

Example:  $45 \bmod 2 = 1 \rightarrow \text{odd}$ ;  $44 \bmod 2 = 0 \rightarrow \text{even}$ .

- Additional applications (e.g., cyclic counters, hash functions) will appear in later lessons.

## Operator Precedence

**Rule** – In JavaScript (and standard arithmetic), **multiplication** and **division** are performed **before** addition and subtraction, unless overridden by parentheses.

- Example expression:  $3 + 5 \times 2$ 
  1. Multiply first:  $5 \times 2 = 10$
  2. Add:  $3 + 10 = 13 \rightarrow \text{Result: } 13$
- Using parentheses to change order:  $(3 + 5) \times 2$ 
  1. Add first:  $3 + 5 = 8$
  2. Multiply:  $8 \times 2 = 16 \rightarrow \text{Result: } 16$

## Precedence Summary

| Level       | Operators | Description              |
|-------------|-----------|--------------------------|
| 1 (highest) | * / %     | Multiply, divide, modulo |

|            |                |                      |
|------------|----------------|----------------------|
| 2          | + -            | Add, subtract        |
| 3 (lowest) | Parentheses () | Force explicit order |

## Best Practices with Parentheses 📌

- **Clarity** – Even when precedence already gives the correct result, adding parentheses makes the intended calculation explicit and improves readability.
- **Maintainability** – Future readers (or yourself) can instantly see the intended evaluation order.

## Dog Age to Human Age Converter 🐕➡️👤

**Goal:** Convert a dog's age to the equivalent human age using the formula  $((\text{dogAge} - 2) * 4) + 21$ .

### Steps

1. Prompt the user for the dog's age.
2. Apply the formula with parentheses to enforce the correct order.
3. Alert the calculated human age.

```
// 1. Get dog age from the user
const dogAge = Number(prompt('How old is your dog?'));

// 2. Compute human age with clear precedence
const humanAge = ((dogAge - 2) * 4) + 21;

// 3. Show the result
alert(`Your dog is ${humanAge} years old in human years.`);
```

- **Explanation of the formula:**
  - Subtract 2 from the dog's age.
  - Multiply the result by 4.
  - Add 21 to obtain the final human-age equivalent.